

Truong Son Nguyen

Executive Summary

Senior Full-stack Developer with 14+ years of experience delivering highly scalable platforms across **.NET**, **Python**, and **Node.js**. I specialize in modernizing legacy systems, leading engineering squads, and shipping cloud-ready services on **Azure**, **AWS**, and **GCP**. Adept at leveraging advanced **AI tools**, **LLM prompting**, and sophisticated **RAG** pipelines to dramatically accelerate delivery and pioneer intelligent, curriculum-grounded applications. Recognized with the **Azure AI-102** certification, demonstrating deep domain expertise in designing and deploying enterprise AI solutions.

Technical Skills

- **Languages & Frameworks:** Python, C#, TypeScript, JavaScript, Next.js, React, Node.js, .NET 6, ASP.NET Core, FastAPI
- **AI & Machine Learning:** LangChain, LangGraph, LLMs (OpenAI, Hugging Face), LlamaIndex, Qdrant, ChromaDB, Pinecone
- **Infrastructure & Cloud:** Azure, AWS, GCP, Docker, Apache Airflow, Kafka, RabbitMQ, CI/CD
- **Databases:** SQL Server, PostgreSQL, MySQL, Cosmos DB

Professional Experience

Homework.ai

Senior Full-stack Developer | Ho Chi Minh City, Vietnam | August 2025 – Present - Delivered an advanced **AI-powered Socratic tutoring application** to thousands of students, by architecting a scalable backend with **React**, **TypeScript**, **FastAPI**, and **PostgreSQL**. - Achieved high-fidelity OCR and problem extraction from complex handwriting and math equations, by seamlessly integrating the **GPT-4 Vision** API. - Enabled real-time, curriculum-grounded tutoring with ultra-low latency, by designing a **WebSocket** server and engineering a sophisticated **Retrieval-Augmented Generation (RAG)** pipeline leveraging **Pinecone** and **Apache Airflow**.

Open Source

Creator / Software Engineer | May 2026 – Present - Streamlined automated profile updates for global developers, by engineering an open-source **FastMCP server** that seamlessly integrates **AI agents** with LinkedIn. - Bypassed API rate limits and CAPTCHAs with a **100% success rate**, by architecting a secure **Chrome DevTools Protocol (CDP)** automation solution using **Playwright** via local browser connections. - Scaled global distribution and ensured rapid deployment, by packaging pre-compiled wheel binaries and automating workflows via **GitHub Actions**.

Personal Projects

AI Engineer | Ho Chi Minh City, Vietnam | April 2026 – Present - Empowered children with a sophisticated AI language assistant named **LingoSafe Tutor**, by architecting complex orchestration loops in **Python** using **LangChain** and **LangGraph**. - Ensured absolute child safety and data integrity, by integrating real-world APIs with strict pedagogical guardrails and rigorous **PII sanitization**. - Mitigated adversarial inputs and prevented data leakage, by implementing advanced AI safety mechanisms including a **Llama-Guard Jailbreak Classifier** and output guardrails backed by **ChromaDB**.

my-coding-agent (Open Source)

Creator / Automation Engineer | March 2026 – Present - Boosted developer productivity by automating intelligent, keyword-based repository routing and task tracking, by architecting a background **Python** orchestrator leveraging **OpenAI** models to integrate **Trello** and **GitHub**. - Synchronized Pull Request states back to Kanban boards seamlessly, by engineering event-driven automated workflows that dispatch **GitHub Actions** implementation jobs upon ticket creation. - Streamlined backend operations, by designing an event-driven architecture utilizing **python-dotenv**, the **OpenAI API**, and the **github-copilot-sdk**.

Curriculum Plan RAG Project

AI Software Engineer | September 2025 – Present - Delivered high-precision retrieval across multilingual curriculum documents, by engineering a robust **RAG** pipeline utilizing **LlamaIndex**, **Qdrant**, and **Hugging Face** embeddings. - Automated highly scalable, incremental data ingestion workflows, by integrating **Apache Airflow**, JSON checkpointing, and **Azure Blob storage**. - Enriched document chunks with granular educational metadata (grade, subject, chapter), by implementing a structure-aware chunking algorithm featuring advanced heading and section detection. - Provided transparent inline citations and meticulously tracked end-to-end retrieval latency, by integrating a sophisticated answer generation layer utilizing **Langfuse**.

Audiostamps

Full Stack Developer | Ho Chi Minh City, Vietnam | June 2025 – Present - Bridged physical objects and digital memories for end-users, by architecting a full-stack web application that enables in-browser recording and playback via scanned **QR-code** stickers. - Streamlined secure transaction processing and order fulfillment, by engineering a robust e-commerce pipeline integrating **PayPal** checkout with automated IPN webhooks. - Handled asynchronous task execution for dynamic PDF label generation and audio format conversions, by building a highly reliable background worker system using **Bull** and **Redis**.

Vactors

Senior Frontend Developer | Ho Chi Minh City, Vietnam | January 2025 – Present - Optimized SEO, performance, and type safety for a collaborative voice-reading platform, by developing a high-performance front-end architecture utilizing **Next.js 14**, **React 18**, and **TypeScript**. - Guaranteed seamless, uninterrupted audio playback and rapid script loading, by implementing highly efficient server-state management and asynchronous data fetching via **React Query**. - Delivered a polished, studio-grade experience directly in the browser, by engineering dynamic user interfaces that integrate **Material UI (MUI)**, **Emotion**, and **React Bootstrap**.

KMS Technology, Inc.

Senior Software Engineer Technical Lead | Ho Chi Minh City, Vietnam | July 2018 – Present - Drove long-term enterprise value and architectural scalability, by providing key leadership, technical stewardship, and mentorship to engineering teams over an 8-year tenure.

TIKI Corporation

Senior Software Engineer | Ho Chi Minh City, Vietnam | September 2016 – July 2018 - Enhanced core systems for Order Fulfillment and Inventory Management on a massive scale, by spearheading development across a comprehensive E-Commerce Platform. - Streamlined invoice processing and merchandise return pipelines, by engineering robust promotion management UI tools and accounting workflows leveraging **Odoo ERP**. - Ensured scalable, maintainable solutions and proactive incident resolution, by authoring detailed technical designs and maintaining rigorous unit testing standards.

NashTech Limited

Back End Developer | Ho Chi Minh City, Vietnam | September 2014 – September 2016 - Empowered customers to seamlessly manage card balances without retailers losing financial control, by engineering a comprehensive Gift Card Management System using **Python, Django, and MySQL**.

EyePower Games Pte Ltd

Senior Software Engineer | Ho Chi Minh City, Vietnam | July 2012 – September 2015 - Drove deep engagement in children's reading, by architecting and developing **Moo-O**, an innovative educational platform leveraging webcams and microphones to transform users into story characters. - Provided students with an accessible, interactive platform to practice language skills, by engineering a scalable web portal for **NewsMaker**. - Boosted educational game performance and reliability, by developing complex backend systems and interactive frontends utilizing **.NET** and **ASP.NET MVC**.

Gold Start Soft

Developer | Ho Chi Minh City, Vietnam | September 2009 – July 2012 - Elevated product visibility and maximized client profitability, by consulting with customers and delivering tailored business portfolio websites utilizing **.NET 3.0, ASP.NET MVC, and SQL Server**.

Education & Certifications

- Artificial Intelligence on Microsoft Azure
- Generative AI: Prompt Engineering Basics
- AI Agents and Agentic AI with Python & Generative AI
- Microsoft Certified: Azure AI Fundamentals
- Microsoft Certified: Azure AI Engineer Associate